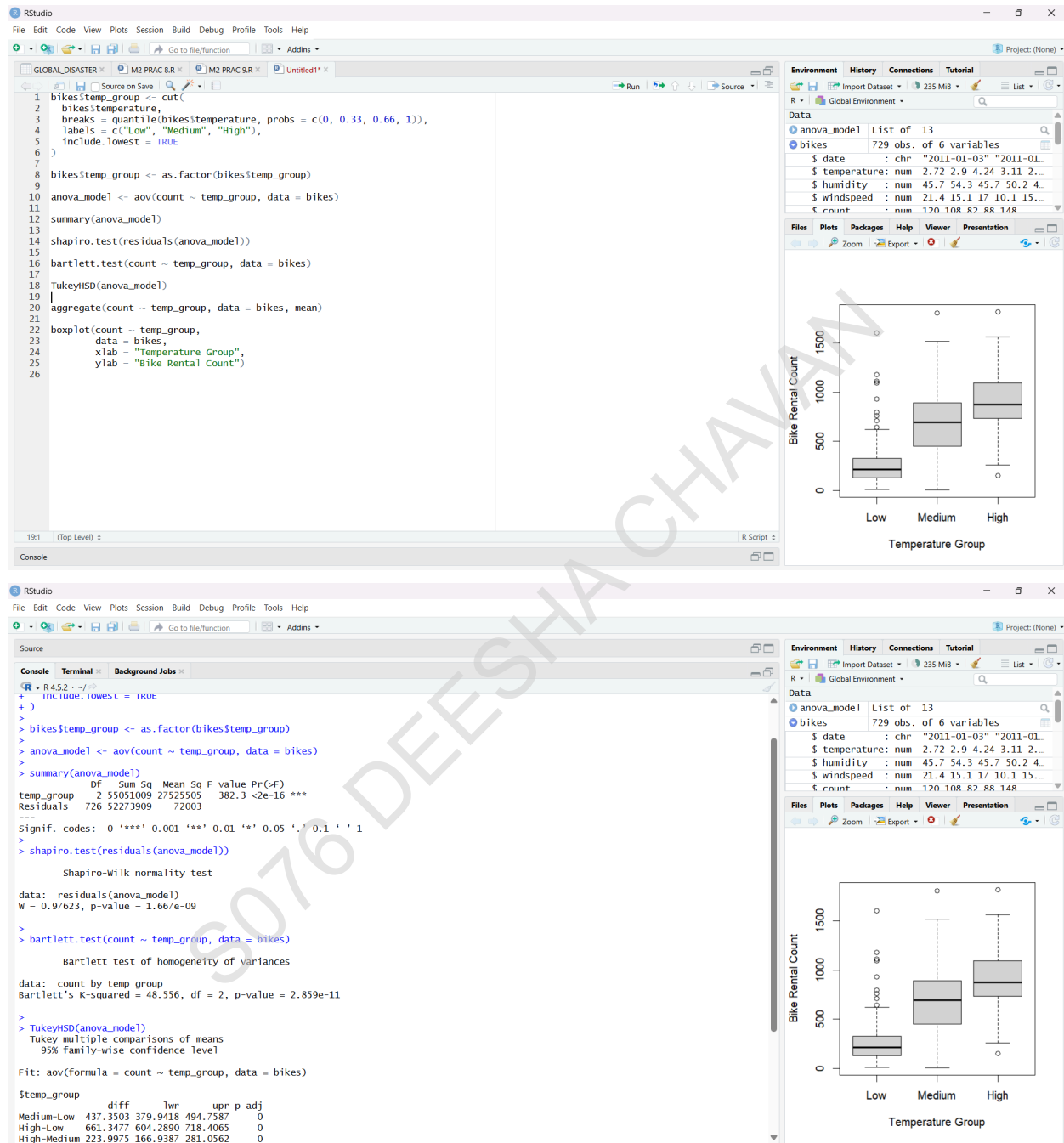


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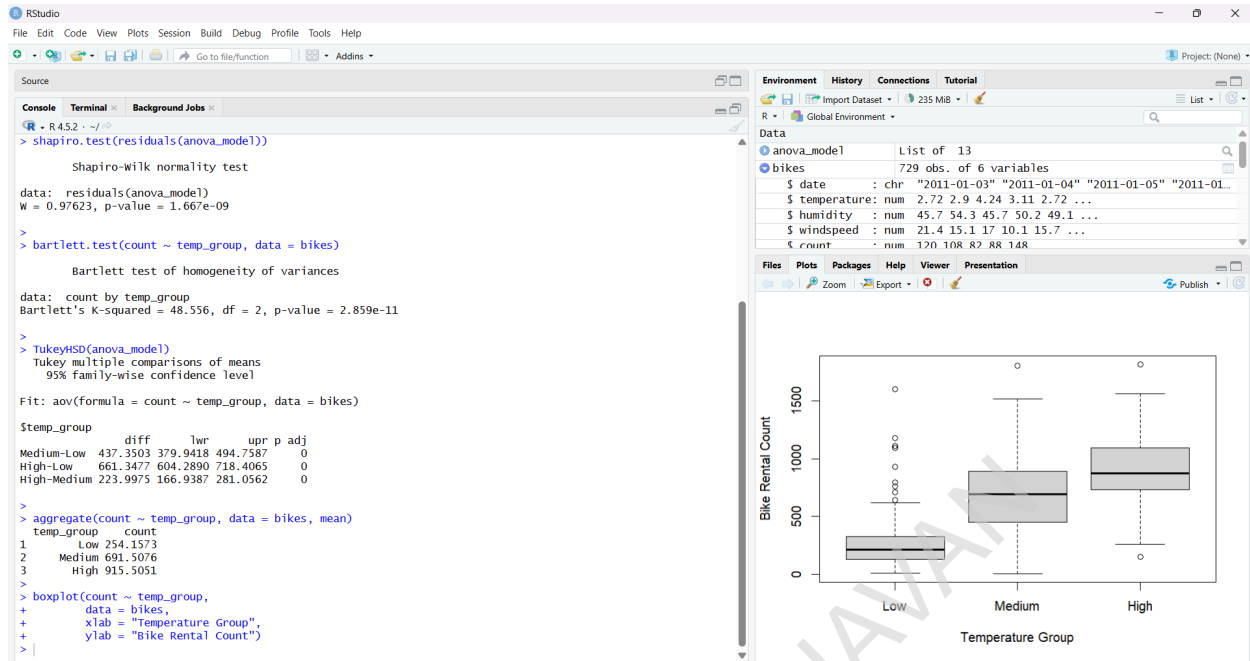
MOD 2 PRACTICAL NO.: 7 - 9

AIM: 7. Performing one-way ANOVA using aov() (R).

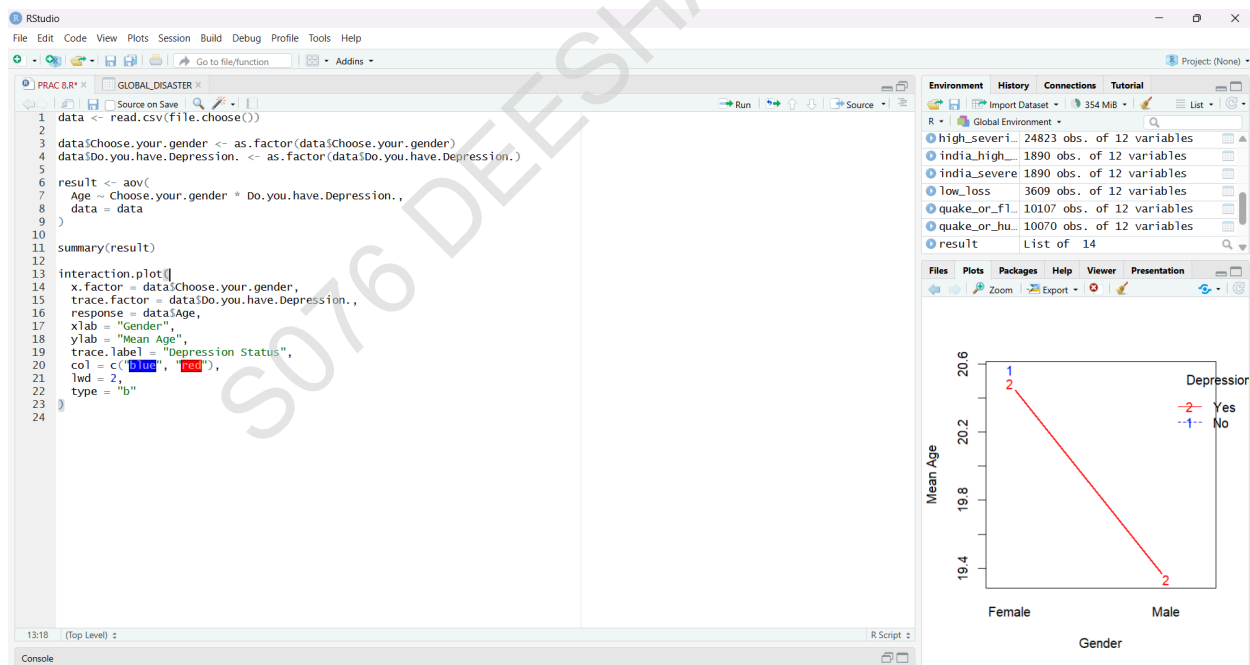


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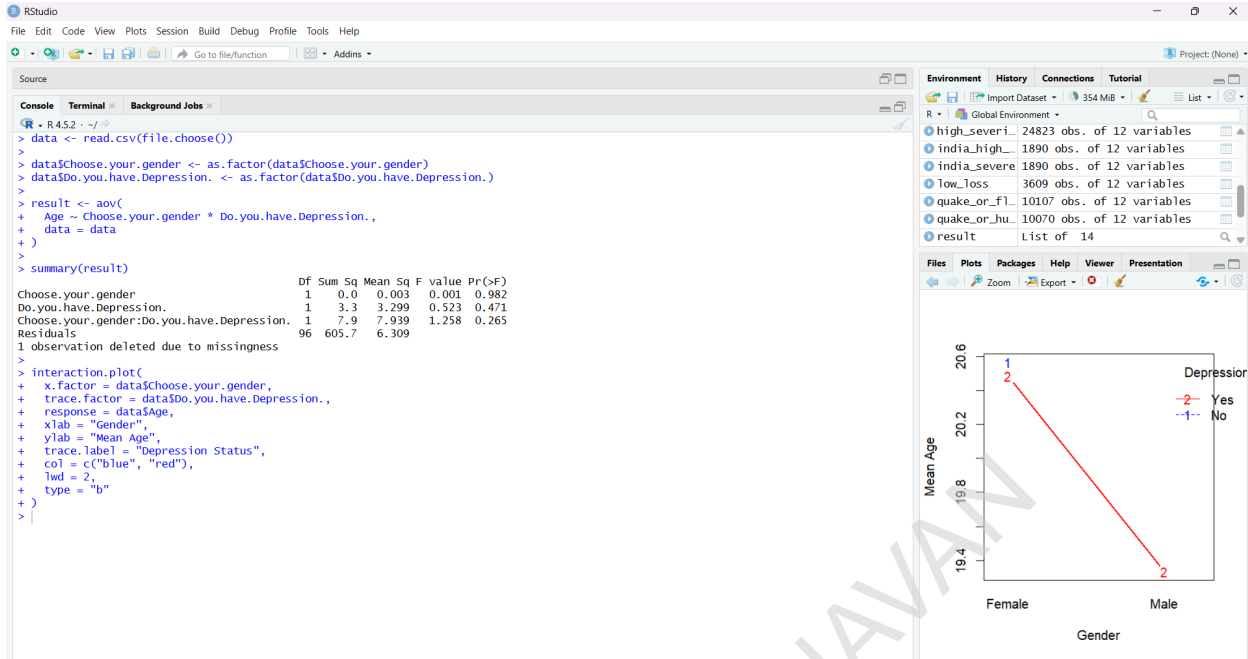
AIM: 8. Performing two-way ANOVA using aov() (R).



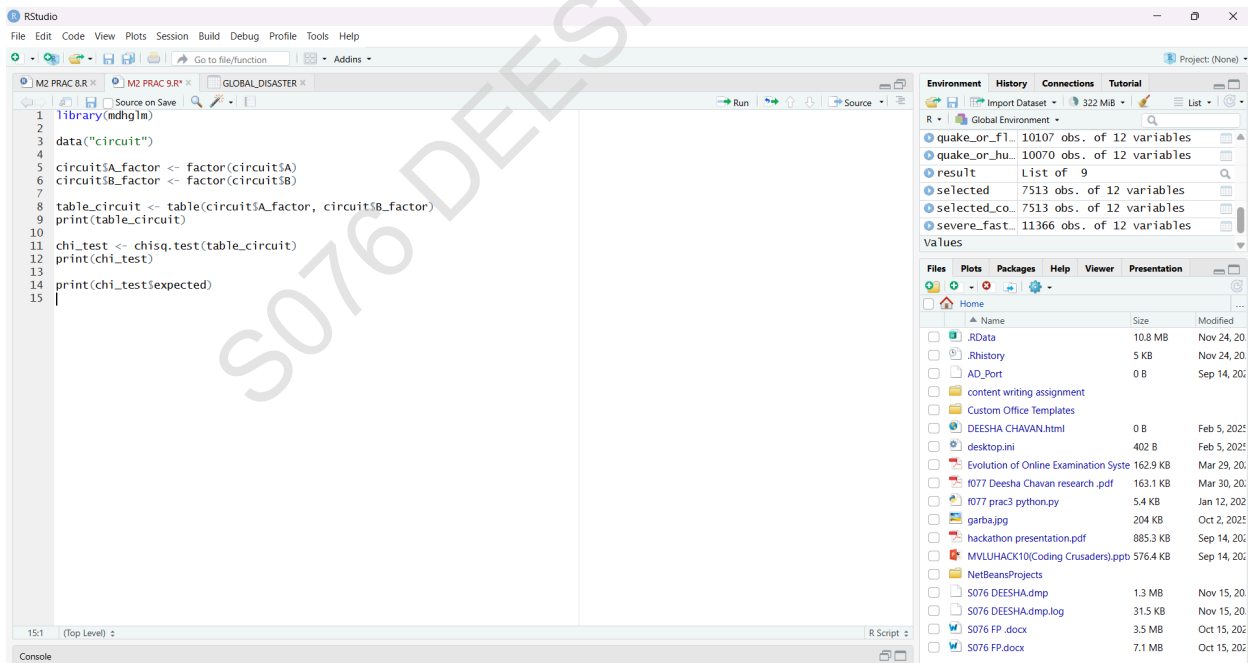
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AIM: 9. Conducting Chi-square tests using `chisq.test()` (R)



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The screenshot shows the RStudio interface. The console window displays the following code and output:

```
> library(mdhglm)
> data("circuit")
> circuit$a_factor <- factor(circuit$a)
> circuit$b_factor <- factor(circuit$b)
> table_circuit <- table(circuit$a_factor, circuit$b_factor)
> print(table_circuit)

  1  2
1 60 25
2 55 25
>
> chi_test <- chisq.test(table_circuit)
> print(chi_test)

Pearson's Chi-squared test with Yates' continuity correction

data: table_circuit
X-squared = 0.0076223, df = 1, p-value = 0.9304
>
> print(chi_test$expected)

      1      2
1 59.24242 25.75758
2 55.75758 24.24242
> |
```

The Environment pane on the right shows the following objects:

- Global Environment
- quake_or_f1... 10107 obs. of 12 variables
- quake_or_hu... 10070 obs. of 12 variables
- result List of 9
- selected 7513 obs. of 12 variables
- selected_co... 7513 obs. of 12 variables
- severe_fast... 11366 obs. of 12 variables

- Independence test, goodness of fit test, homogeneity test

The screenshot shows the RStudio interface. The console window displays the following code and output:

```
1 # create contingency table
2 data <- matrix(c(30, 20, 10, 40),
3               nrow = 2,
4               byrow = TRUE)
5
6 rownames(data) <- c("Male", "Female")
7 colnames(data) <- c("Like", "Dislike")
8
9 # chi-square test of independence
10 chisq.test(data)
11
12 |
```

The console output shows:

```
> Create contingency table
> data <- matrix(c(30, 20, 10, 40),
+               nrow = 2,
+               byrow = TRUE)
>
> rownames(data) <- c("Male", "Female")
> colnames(data) <- c("Like", "Dislike")
>
> # Chi-square test of independence
> chisq.test(data)

Pearson's Chi-squared test with Yates' continuity correction

data: data
X-squared = 15.042, df = 1, p-value = 0.0001052
> |
```

The Environment pane on the right shows the following objects:

- Global Environment
- BostonHousi... 1 obs. of 1 variable
- BostonHousi... 506 obs. of 14 variables
- num [1:2, 1:2] 30 10 20 40
- global_afr... 23463 obs. of 16 variables

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This screenshot shows the RStudio interface with a script editor, console, and environment pane. The script editor contains R code for creating a contingency table and performing a chi-square test of independence. The console shows the execution of the code, and the environment pane displays the objects created.

```
1 # create contingency table
2 data <- matrix(c(30, 20, 10, 40),
3               nrow = 2,
4               byrow = TRUE)
5
6 rownames(data) <- c("Male", "Female")
7 colnames(data) <- c("Like", "Dislike")
8
9 # chi-square test of independence
10 chisq.test(data)
11
12 # observed frequencies
13 observed <- c(40, 35, 25)
14
15 # Expected probabilities
16 expected_prob <- c(1/3, 1/3, 1/3)
17
18 # Goodness of fit test
19 chisq.test(observed, p = expected_prob)
20
```

Console output:

```
> # observed frequencies
> observed <- c(40, 35, 25)
>
> # Expected probabilities
> expected_prob <- c(1/3, 1/3, 1/3)
>
> # Goodness of fit test
> chisq.test(observed, p = expected_prob)

Chi-squared test for given probabilities

data: observed
X-squared = 3.5, df = 2, p-value = 0.1738
> |
```

Environment pane:

Object	Class	Attributes
BostonHousi...	num	1 obs. of 1 variable
BostonHousi...	num	506 obs. of 14 variables
data	num	[1:2, 1:2] 30 10 20 40
global_air_...	num	23463 obs. of 16 variables

This screenshot shows the RStudio interface with a script editor, console, and environment pane. The script editor contains R code for creating a data matrix and performing a chi-square test of homogeneity. The console shows the execution of the code, and the environment pane displays the objects created.

```
10 chisq.test(data)
11
12 # observed frequencies
13 observed <- c(40, 35, 25)
14
15 # Expected probabilities
16 expected_prob <- c(1/3, 1/3, 1/3)
17
18 # Goodness of fit test
19 chisq.test(observed, p = expected_prob)
20
21 # Create data matrix
22 data <- matrix(c(20, 30,
23               25, 35,
24               30, 40),
25               nrow = 3,
26               byrow = TRUE)
27
28 rownames(data) <- c("Group A", "Group B", "Group C")
29 colnames(data) <- c("Yes", "No")
30
31 # chi-square test of homogeneity
32 chisq.test(data)
33
34
```

Console output:

```
> + 30, 40),
> + 25, 35,
> + 30, 40),
> nrow = 3,
> byrow = TRUE)
>
> rownames(data) <- c("Group A", "Group B", "Group C")
> colnames(data) <- c("Yes", "No")
>
> # chi-square test of homogeneity
> chisq.test(data)

Pearson's Chi-squared test

data: data
X-squared = 0.097959, df = 2, p-value = 0.9522
> |
```

Environment pane:

Object	Class	Attributes
BostonHousi...	num	1 obs. of 1 variable
BostonHousi...	num	506 obs. of 14 variables
data	num	[1:3, 1:2] 20 25 30 30 3...
global_air_...	num	23463 obs. of 16 variables

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